

CTMT Covariance Geometry

Resolution-Relative Covariance, Coupling Rigidity, and a Rigorous Account of Singular
Covariance

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Abstract

This paper develops the covariance-geometry layer of Compatible Transport Morphing Theory (CTMT), companion to the resolution-geometry foundation $\mathcal{G} = (O, g; R, \Sigma)$. Given an observation space O with a fixed reference metric g , a response Jacobian J with resolved sector $R = \text{Im}(J)$ and null sector $N = R^\perp_g$, and an observation covariance Σ , we study the resolution-relative object

$$\text{CovGeom}(J, \Sigma) := (R, N, \Sigma_R, \Sigma_N, C_{RN}).$$

We do not claim a privileged covariance estimator; standard statistics supplies Σ , and CTMT supplies the geometry in which a declared estimate is decomposed, transported, compared, and falsified. We prove: (i) that the Fisher pullback $J^\top \Sigma^{-1} J$ equals the inverse Schur complement $J_R^\top (\Sigma_R - C_{RN} \Sigma_N^{-1} C_{RN}^\top)^{-1} J_R$ and is therefore blind to Σ_N when $C_{RN} = 0$ and many-to-one in general (precise Fisher insufficiency); (ii) that in the whitening geometry $g = \Sigma^{-1}$ covariance preservation coincides with g -isometry, unifying the covariance automorphism group with the admissible automorphism group of the foundation; and (iii) a complete treatment of *singular* covariance through the support/deterministic splitting, with the exact criterion $R \subseteq \text{range}(\Sigma)$ under which the generalized Fisher form $J^\top \Sigma^+ J$ equals the true (degenerate-Gaussian) Fisher information, and the earlier null-supported result recovered as an extreme corner. The paper closes with resolution-relative similarity diagnostics, a conservative estimation protocol, falsification criteria, and non-claims.

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1 Purpose and claim stratification

The local CTMT morphism problem is organised by the resolution geometry $\mathcal{G} = (O, g; R, \Sigma)$ of the foundation paper. The present paper answers a narrower question:

Once the resolved/null decomposition is fixed, what is covariance as a CTMT geometric object, and what does that structure determine and forbid?

The answer is *not* that CTMT owns a universal estimator for Σ . Statistics already supplies many (sample, shrinkage, robust, Bayesian, state-space, instrumental). CTMT enters afterwards: it decomposes the declared covariance relative to the resolution structure and tests admissibility under transport. We separate four layers, and are explicit about which does what.

Statistical.

A declared protocol produces a covariance estimate or model $\Sigma \in \text{PSD}(O)$.

Resolution.

The response Jacobian J determines $R = \text{Im}(J)$ and, via the fixed metric g , the null sector $N = R^{\perp_g}$.

Covariance-geometry.

Relative to $O = R \oplus N$, Σ decomposes into the operators $\Sigma_R, \Sigma_N, C_{RN}$.

Morphism.

Admissible local morphisms are those preserving the declared resolution geometry (strictly, or conformally when the protocol discards absolute covariance scale).

Remark 1.1. This paper is the intermediate layer between the covariance estimate and the automorphism classification. It neither replaces statistical covariance estimation nor claims that J alone determines Σ .

2 Chart, sectors, and resolution-relative covariance

We use the conventions of the foundation. In particular the observation space carries a fixed reference metric that makes the null sector well defined; this is the single most important alignment relative to the previous draft.

Assumption 2.1 (Observation space and reference metric). The observation space is a finite-dimensional real inner-product space (O, g) , $g(u, v) = u^\top G v$ with $G = G^\top \succ 0$. When Σ is nonsingular the canonical choice is the whitening metric $G = \Sigma^{-1}$; when Σ is singular whitening is unavailable and a fixed units metric G is used, with Σ handled through its pseudo-inverse (Section 6). All orthogonality, adjoints, and projectors are taken with respect to g .

Definition 2.2 (Operational observation chart). An operational observation chart is a quadruple $\mathcal{C} = (O, \Theta, F, \Sigma)$, where O is the observation space, Θ the space of admissible perturbations, $F : \Theta \rightarrow O$ the response map, and $\Sigma \in \text{PSD}(O)$ the declared covariance. The response Jacobian at θ_0 is $J = D_{\theta}F(\theta_0) : T_{\theta_0}\Theta \rightarrow O$.

Definition 2.3 (Sectors and projectors). The resolved and null sectors are

$$R := \text{Im}(J) \subseteq O, \quad N := R^{\perp_g} \subseteq O,$$

so $O = R \oplus N$ (g -orthogonally). We write Π_R, Π_N for the g -orthogonal projectors; in a g -orthonormal frame $N = \text{Ker}(J^*)$ and $\Pi_R = JJ^+$, $\Pi_N = I - JJ^+$ with J^+ the Moore–Penrose pseudo-inverse.

Remark 2.4 (Null $\neq$ physically absent). N is not a claim that unresolved directions do not exist; it is the set of directions not activated at first order by admissible perturbations. Null covariance may be operationally meaningful even where the Fisher pullback cannot see it (Proposition 3.2).

Definition 2.5 (Strict and conformal covariance structure). A *strict* covariance structure is a form $\Sigma \in \text{PSD}(O)$ whose absolute scale is part of the protocol; when $\Sigma \succ 0$ we write $\Sigma \in \text{SPD}(O)$. A *conformal* covariance structure is a ray $[\Sigma] = \{\lambda\Sigma : \lambda > 0\}$, used only when the protocol explicitly quotients out global covariance scale.

Definition 2.6 (Resolution-relative blocks and covariance geometry). The (basis-free) blocks are the operators

$$\Sigma_R := \Pi_R \Sigma \Pi_R, \quad \Sigma_N := \Pi_N \Sigma \Pi_N, \quad C_{RN} := \Pi_R \Sigma \Pi_N,$$

and the CTMT covariance geometry of the chart is

$$\text{CovGeom}(J, \Sigma) := (R, N, \Sigma_R, \Sigma_N, C_{RN}),$$

displayed in an adapted g -orthonormal frame as $\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^* & \Sigma_N \end{pmatrix}$. In the conformal case one retains the blocks up to one common positive scale.

Proposition 2.7 (Existence and uniqueness of blocks). *For a fixed decomposition $O = R \oplus N$ and any $\Sigma \in \text{PSD}(O)$, the block data $(\Sigma_R, \Sigma_N, C_{RN})$ exists and is unique.*

Proof. Each $x \in O$ decomposes uniquely as $x = \Pi_R x + \Pi_N x$; the bilinear form Σ is thus determined by its restrictions to $R \times R$, $N \times N$, $R \times N$, which are exactly the three operators. $\square$

The cross block is not a nuisance term: by Theorem 5.2 it changes the automorphism group and hence the admissible morphism class.

3 Fisher information is insufficient (exactly)

We make precise, and prove exactly, the claim that Fisher geometry does not determine covariance geometry. The tool is the Schur complement; the identity below is verified numerically to machine precision.

Definition 3.1 (Fisher / response pullback). When $\Sigma \succ 0$, the Fisher pullback (response metric) on Θ is $G_F := J^\top \Sigma^{-1} J$.

Proposition 3.2 (Fisher equals the inverse Schur complement). *Work in an adapted g -orthonormal frame and write $J = \begin{pmatrix} J_R \\ 0 \end{pmatrix}$ (its image is R). Then*

$$G_F = J^\top \Sigma^{-1} J = J_R^\top (\Sigma_R - C_{RN} \Sigma_N^{-1} C_{RN}^\top)^{-1} J_R = J_R^\top (\Sigma / \Sigma_N)^{-1} J_R,$$

where Σ / Σ_N is the Schur complement of Σ_N in Σ . Consequently:

1. If $C_{RN} = 0$, then $G_F = J_R^\top \Sigma_R^{-1} J_R$ depends only on Σ_R and is completely blind to Σ_N .
 2. In general the map $(\Sigma_R, \Sigma_N, C_{RN}) \mapsto G_F$ is many-to-one: any two covariance geometries with the same Schur complement and the same J yield identical Fisher pullback.
- Hence G_F does not determine $\text{CovGeom}(J, \Sigma) = (R, N, \Sigma_R, \Sigma_N, C_{RN})$.

Proof. Since $\text{Im}(J) = R$, $J^\top \Sigma^{-1} J = J_R^\top (\Sigma^{-1})_{RR} J_R$, where $(\Sigma^{-1})_{RR}$ is the R -block of the full inverse. By the block matrix inversion formula, $(\Sigma^{-1})_{RR} = (\Sigma_R - C_{RN} \Sigma_N^{-1} C_{RN}^\top)^{-1}$, the inverse Schur complement. Both consequences are immediate: setting $C_{RN} = 0$ gives $\Sigma/\Sigma_N = \Sigma_R$; and any perturbation of (Σ_N, C_{RN}) preserving $C_{RN} \Sigma_N^{-1} C_{RN}^\top$ leaves G_F fixed. $\square$

Remark 3.3. This is the rigorous form of “Fisher insufficiency.” It is not a slogan: the Fisher pullback reads covariance only through the resolved response and only through one specific combination of the blocks. Null covariance Σ_N is invisible to it whenever the sectors decouple, and the coupling C_{RN} is recoverable from G_F only jointly with Σ_N , never on its own.

4 Unification with the resolution-geometry automorphisms

The previous draft defined “strict covariance-geometry automorphisms” by $\phi(R) = R$, $\phi(N) = N$, $\phi \Sigma \phi^\top = \Sigma$, and asserted $\phi(N) = N'$ in transport without proof. Both are cleaned up by the following observation, which also identifies this paper’s morphisms with the foundation’s.

Proposition 4.1 (Whitening turns covariance preservation into isometry). *Let $\Sigma \succ 0$ and take the whitening metric $g = \Sigma^{-1}$ (and $g' = \Sigma'^{-1}$). Then a linear isomorphism $\phi : O \rightarrow O'$ satisfies the covariance push-forward law $\phi \Sigma \phi^\top = \Sigma'$ if and only if ϕ is a (g, g') -isometry. In that case $\phi(N) = N'$ automatically.*

Proof. A (g, g') -isometry satisfies $\phi^\top G' \phi = G$, i.e. $\phi^\top \Sigma'^{-1} \phi = \Sigma^{-1}$, equivalently $\Sigma' = \phi \Sigma \phi^\top$. Given the isometry and $\phi(R) = R'$, $\phi(N) = \phi(R^\perp_g) = (\phi R)^\perp_{g'} = R'^\perp_{g'} = N'$. $\square$

Corollary 4.2 (The two automorphism groups coincide). *For $\Sigma \succ 0$ with $g = \Sigma^{-1}$, the covariance-geometry automorphism group of Definition 5.1 equals the admissible automorphism group $\text{Aut}(\mathcal{G})$ of the foundation. The covariance layer is therefore not a separate theory but the concrete realisation of the foundation’s automorphisms in the whitening frame.*

5 Automorphism structure and coupling rigidity

Definition 5.1 (Covariance-geometry automorphism). An automorphism of $\text{CovGeom}(J, \Sigma)$ is a g -isometry ϕ of O with $\phi(R) = R$ and $\phi \Sigma \phi^\top = \Sigma$. (By Lemma-level reasoning $\phi(N) = N$ follows.)

Theorem 5.2 (Automorphisms are coupling stabilizers). *Every automorphism is g -orthogonally block diagonal, $\phi = \text{diag}(A, B)$ with $A \in \text{O}(g|_R)$, $B \in \text{O}(g|_N)$, and*

$$\text{Aut}(\text{CovGeom}) = \{ \text{diag}(A, B) : A \Sigma_R A^* = \Sigma_R, B \Sigma_N B^* = \Sigma_N, A C_{RN} B^* = C_{RN} \}.$$

In particular:

1. Decoupled. If $C_{RN} = 0$ then $\text{Aut}(\text{CovGeom}) = \text{Aut}(R; \Sigma_R) \times \text{Aut}(N; \Sigma_N)$, and after whitening this is $\text{O}(r) \times \text{O}(m - r)$ with $r = \text{rank } J$, $m = \dim O$.
2. Coupled. If $C_{RN} \neq 0$ then $\text{Aut}(\text{CovGeom})$ is the stabilizer of C_{RN} inside $\text{Aut}(R; \Sigma_R) \times \text{Aut}(N; \Sigma_N)$, a proper closed subgroup. Resolved-null coupling is thus a morphism-class invariant and the exact source of rigidity.

Here $\text{Aut}(R; \Sigma_R) = \{ A \in \text{O}(g|_R) : A \Sigma_R A^* = \Sigma_R \}$, and likewise on N .

Proof. Preserving the complementary subspaces R, N forces block-diagonal form; the g -isometry condition puts A, B in the sector orthogonal groups. Writing $\phi\Sigma\phi^\top$ in blocks and equating to Σ gives the three displayed equations. The consequences are the vanishing (resp. non-vanishing) of the third constraint, exactly as in the foundation's structure theorem. $\square$

This coincides, as it must by Corollary 4.2, with the automorphism structure theorem of the foundation paper.

Theorem 5.3 (Conformal automorphisms). *For the conformal geometry $(R, N, [\Sigma])$,*

$$\text{Aut}_c = \{\text{diag}(A, B) : \exists \lambda > 0, A\Sigma_R A^* = \lambda\Sigma_R, B\Sigma_N B^* = \lambda\Sigma_N, AC_{RN}B^* = \lambda C_{RN}\},$$

and if $C_{RN} = 0$ then, after whitening, $\text{Aut}_c \cong \mathbb{R}_+ \times \text{O}(r) \times \text{O}(m-r)$. Independent sector rescalings are not conformal equivalences unless a separate sector-relative protocol is declared.

6 Singular covariance

We now answer the question the previous draft left open: what does the covariance geometry say when Σ is *singular*? Whitening is then unavailable and the Fisher pullback $J^\top \Sigma^{-1} J$ is undefined, so a different, honest device is needed. The device is the Moore–Penrose pseudo-inverse and the splitting of O by the covariance support. Everything below is standard degenerate-Gaussian and generalized-inverse theory; the finite-Fisher criterion and the identity $G_F = J^\top \Sigma^+ J$ are verified numerically.

6.1 Support and deterministic subspaces

Definition 6.1 (Support and deterministic subspace). For $\Sigma \in \text{PSD}(O)$ (with a fixed units metric g), the *observational support* and the *deterministic subspace* are

$$S := \text{range}(\Sigma) \subseteq O, \quad D := \text{Ker}(\Sigma) = S^{\perp_g},$$

with g -orthogonal projector Π_S . Let Σ^+ be the (g -)Moore–Penrose pseudo-inverse, so $\Sigma\Sigma^+ = \Sigma^+\Sigma = \Pi_S$.

The deterministic subspace D is the set of observable directions carrying *no* observational uncertainty: components of y along D are measured exactly. Singular covariance is precisely the presence of such noise-free directions, $D \neq \{0\}$.

Definition 6.2 (Degeneracy type). The interaction of the resolution splitting $O = R \oplus N$ with the covariance splitting $O = S \oplus D$ classifies the chart. The four invariant intersections

$$R \cap S, \quad R \cap D, \quad N \cap S, \quad N \cap D$$

are, respectively: resolved-noisy, resolved-deterministic, unresolved-noisy (seepage), and unresolved-deterministic directions.

6.2 Generalized Fisher form and the finite-information criterion

Definition 6.3 (Generalized Fisher form). For possibly singular Σ , the generalized Fisher (response) form is $\widetilde{G}_F := J^\top \Sigma^+ J$, always defined.

Theorem 6.4 (When the generalized form is the true Fisher information). *Let $y \sim \mathcal{N}(f(\theta), \Sigma)$ be a (possibly degenerate) Gaussian observation model with parameter-independent Σ and $J = \partial f / \partial \theta$.*

1. If $R \subseteq S$ (equivalently $R \cap D = \{0\}$: no resolved deterministic direction), then the Fisher information matrix is finite and equals the generalized form,

$$\mathcal{I}(\theta) = J^\top \Sigma^+ J = \tilde{G}_F.$$

2. If $R \not\subseteq S$ (equivalently $R \cap D \neq \{0\}$), then along $R \cap D$ the observation is noise-free, the corresponding parameter combination is locally determined exactly, and the Fisher information is unbounded in those directions; the generalized form $\tilde{G}_F$ remains finite and underestimates the true information.

Proof. A degenerate Gaussian is supported on the affine subspace $f(\theta) + S$; its log-density there is $-\frac{1}{2}(y - f)^\top \Sigma^+ (y - f)$ up to a constant. If $R = \text{Im}(J) \subseteq S$, then as θ varies the mean stays within a single support subspace, the score is $J^\top \Sigma^+ (y - f)$, and

$$\mathcal{I} = \mathbb{E}[J^\top \Sigma^+ (y - f)(y - f)^\top \Sigma^+ J] = J^\top \Sigma^+ \Sigma \Sigma^+ J = J^\top \Sigma^+ J,$$

using $\Sigma^+ \Sigma \Sigma^+ = \Sigma^+$. If instead some $Jv \in R \cap D$, then varying θ along v moves the mean along a zero-variance direction: distinct parameter values give mutually singular observation laws, so the parameter is locally identified exactly and $\mathcal{I}$ is unbounded along v , while $J^\top \Sigma^+ J$ stays finite because Σ^+ annihilates D . $\square$

Corollary 6.5 (Operational meaning of singular covariance). *The criterion $R \subseteq \text{range}(\Sigma)$ is exactly the condition “no resolved observable is noise-free.” Under it, all Fisher and response-metric statements of the foundation carry over verbatim with Σ^{-1} replaced by Σ^+ . A violation is not a defect of the theory but a detectable feature: a resolved deterministic direction is an exact first-order constraint on the parameters, to be reported and (numerically) deflated rather than inverted.*

6.3 Null-supported covariance as an extreme corner

The previous draft’s “canonical support of null-admissible covariance” is the maximally singular resolved case, recovered here as a corollary.

Proposition 6.6 (Null-supported covariance). *Suppose the covariance is supported in the null sector, $S \subseteq N$ (equivalently $R \subseteq D$: every resolved direction is noise-free). Then*

$$\Sigma = \Pi_N \Sigma \Pi_N, \quad \text{range}(\Sigma) \subseteq N, \quad C_{RN} = 0, \quad \tilde{G}_F = J^\top \Sigma^+ J = 0.$$

Thus null-supported covariance carries no first-order resolved information: the resolved sector is measured exactly and all observational uncertainty lives in the unresolved directions.

Proof. $S \subseteq N$ gives $\Sigma = \Pi_N \Sigma \Pi_N$ and $\text{range}(\Sigma) \subseteq N$, whence $C_{RN} = \Pi_R \Sigma \Pi_N = 0$. Since Σ^+ is supported on $S \subseteq N$ and $\text{Im}(J) = R \perp_g N$, we have $\Sigma^+ J = 0$, so $\tilde{G}_F = 0$. $\square$

Remark 6.7 (Where whitening must be abandoned). Singular covariance is exactly the regime where the canonical whitening metric $g = \Sigma^{-1}$ does not exist. The support/pseudo-inverse treatment above is the price of that: one fixes a units metric g , works with Σ^+ on S , and reports $D = \text{Ker } \Sigma$ explicitly. This is the honest boundary of the whitening geometry, and it is where the two-metric bookkeeping of the foundation earns its keep.

7 Transport, similarity, and estimation

7.1 Transport

Definition 7.1 (Strict / conformal chart isomorphism). A strict chart isomorphism is a pair (ϕ, a) , $\phi : O \rightarrow O'$ a (g, g') -isometry and $a : T\Theta \rightarrow T\Theta'$ invertible, with $J'a = \phi J$ and $\phi \Sigma \phi^\top = \Sigma'$. It is conformal if the last equality is relaxed to $\phi \Sigma \phi^\top = \lambda \Sigma'$, $\lambda > 0$, admissible only when the protocol discards absolute scale.

Proposition 7.2 (Functoriality). *A strict chart isomorphism satisfies $\phi(R) = R'$, $\phi(N) = N'$, and carries $(\Sigma_R, \Sigma_N, C_{RN})$ to $(\Sigma'_R, \Sigma'_N, C'_{RN})$; hence it preserves CTMT covariance geometry. Under a conformal isomorphism all three blocks scale by the same λ .*

Proof. $J'a = \phi J$ with a invertible gives $\text{Im}(J') = \phi(\text{Im}(J))$, i.e. $\phi(R) = R'$; since ϕ is a g -isometry, $\phi(N) = N'$ by Proposition 4.1. Restricting $\phi \Sigma \phi^\top = \Sigma'$ to the sector pairs preserves the three blocks. The conformal case scales the right-hand side by λ . $\square$

The requirement that ϕ be a g -isometry (added here) is what makes $\phi(N) = N'$ a theorem rather than an assertion; by Proposition 4.1 it is automatic in the whitening frame.

7.2 Similarity (a declared diagnostic, not an invariant metric)

Similarity of two covariance geometries is compared *structure by structure*, not by Fisher metrics or raw covariances alone. With subspace distances $d_R = \|\Pi_{R_1} - \Pi_{R_2}\|_F$, $d_N = \|\Pi_{N_1} - \Pi_{N_2}\|_F$ and, on SPD blocks, the affine-invariant distance $d_{\text{SPD}}(A, B) = \|\log(A^{-1/2}BA^{-1/2})\|_F$, a conservative composite is

$$d_{\text{CTMT}} = \alpha_R d_R + \alpha_N d_N + \beta_R d_{\text{SPD}}(\Sigma_{R,1}, \Sigma_{R,2}) + \beta_N d_{\text{SPD}}(\Sigma_{N,1}, \Sigma_{N,2}) + \gamma \frac{\|C_1 - C_2\|_F}{\max(\|\Sigma_1\|_F, \|\Sigma_2\|_F, \varepsilon)}.$$

The weights must be preregistered, not tuned to a desired conclusion. This is a comparison diagnostic; the invariant content being compared is $(R, N, \Sigma_R, \Sigma_N, C_{RN})$, and, for singular blocks, the regularized or pseudo-inverse variants of d_{SPD} .

7.3 Estimation protocol

1. Declare the observation vector and preprocessing; fix the units metric g .
2. Estimate $\widehat{\Sigma}$ by an established method (sample, shrinkage, robust, instrumental). Report rank $\widehat{\Sigma}$ and the deterministic subspace $\widehat{D} = \text{Ker } \widehat{\Sigma}$ under a declared tolerance.
3. Estimate J ; report $r = \text{rank } J$, conditioning, and sensitivity to the rank threshold.
4. Compute $R = \text{Im}(J)$, $N = R^\perp_g$, and the g -orthogonal projectors.
5. Choose the working metric: whitening $g = \widehat{\Sigma}^{-1}$ if $\widehat{\Sigma}$ is well conditioned; otherwise keep the units metric and use $\widehat{\Sigma}^+$.
6. Block-decompose $\widehat{\Sigma}$ into $\widehat{\Sigma}_R, \widehat{\Sigma}_N, \widehat{C}_{RN}$.
7. Report the normalized coupling $\kappa_{RN} = \|\widehat{C}_{RN}\|_F / (\|\widehat{\Sigma}\|_F + \varepsilon)$ and the resolved-deterministic dimension $\dim(R \cap \widehat{D})$.
8. Select strict or conformal interpretation per protocol.
9. Test a candidate ϕ via sector residuals d_R, d_N and the covariance residual $\|\phi \widehat{\Sigma} \phi^\top - \widehat{\Sigma}\|_F / (\|\widehat{\Sigma}\|_F + \varepsilon)$ (or its conformal $\min_{\lambda > 0}$ variant).
10. Report accept/reject against preregistered gates.

8 Falsification criteria

Falsifier 8.1 (Rank instability). If rank J is unstable under admissible perturbations, windows, or thresholds, the resolved/null decomposition is unreliable for that chart.

Falsifier 8.2 (Null leakage). For a claimed null-supported covariance C , large persistent leakage $\ell = (\|\Pi_R C \Pi_R\|_F + \|\Pi_R C \Pi_N\|_F) / (\|C\|_F + \varepsilon)$ refutes the null-support hypothesis (Proposition 6.6).

Falsifier 8.3 (Cross-block instability). If κ_{RN} varies erratically across stabilized windows without protocol explanation, the covariance geometry is not stable.

Falsifier 8.4 (Singular-information direction). If $\dim(R \cap \text{Ker } \Sigma) > 0$, a resolved observable is noise-free: the generalized form $J^\top \Sigma^+ J$ underestimates the true information (Theorem 6.4). Report and deflate the direction; do not invert Σ .

Falsifier 8.5 (Fisher-only false positive). If two charts have similar Fisher pullbacks but different N, Σ_N, C_{RN} , Fisher similarity is a false positive for covariance-geometry similarity (Proposition 3.2).

Falsifier 8.6 (Transport non-composability). If covariance geometry appears preserved on individual transitions but fails around a closed loop, the local similarities do not define admissible transport (nontrivial holonomy, in the foundation’s sense).

9 Limitations and non-claims

Non-claim 9.1 (No universal estimator). CTMT does not replace covariance estimation; it supplies the geometry in which a declared estimate is decomposed, transported, compared, and falsified.

Non-claim 9.2 (Jacobian does not determine full covariance). J determines R, N , and the support constraints on null-admissible covariance; it does not determine the covariance spectrum, which requires data or a declared noise model.

Non-claim 9.3 (No Fisher-only closure). $G_F = J^\top \Sigma^{-1} J$ (or $J^\top \Sigma^+ J$) is a derived diagnostic equal to the inverse Schur complement; by Proposition 3.2 it is insufficient to determine covariance geometry.

Non-claim 9.4 (Singular case is pseudo-inverse, not inverse). For singular Σ the generalized form $J^\top \Sigma^+ J$ equals the true Fisher information only under $R \subseteq \text{range}(\Sigma)$; otherwise it underestimates, and the theory reports the noise-free directions rather than inverting through them.

Non-claim 9.5 (No automatic global theorem). Local covariance-geometry compatibility does not by itself establish global bundle consistency or holonomy closure; those are the separate transport problems of the foundation’s globalization section.

Non-claim 9.6 (No physical universality). No claim is made of a physical law, prediction scheme, Navier–Stokes result, or measurement-theoretic derivation. The paper states a local covariance geometry and its falsifiers.

10 Discussion and conclusion

Once a resolved/null decomposition is fixed, covariance ceases to be “just a noise matrix” or “just a Fisher weight” and becomes the structured object $\Sigma \leftrightarrow (\Sigma_R, \Sigma_N, C_{RN})$. The resolved block records covariance of first-order observables; the null block records covariance invisible to resolved parameter motion; the cross block couples them and, by Theorem 5.2, controls the automorphism group. Fisher geometry sees only the inverse Schur complement, and is therefore a diagnostic, not the primitive.

The genuinely new content of this hardened version is the treatment of *singular* covariance. Singular Σ means noise-free observable directions, $D = \text{Ker } \Sigma \neq \{0\}$. The support/pseudo-inverse formulation gives a complete account: the generalized Fisher form $J^\top \Sigma^+ J$ equals the

true degenerate-Gaussian information exactly when $R \subseteq \text{range}(\Sigma)$, the null-supported case is the extreme corner $R \subseteq D$, and a resolved deterministic direction is a detectable, reportable feature rather than a breakdown. This closes, at the covariance layer, the “degenerate covariance” problem left open in the foundation. Globalization of the singular case (support fields that change rank across charts) remains open and belongs to the transport-holonomy program.

$$\boxed{\text{CovGeom}(J, \Sigma) = (R, N, \Sigma_R, \Sigma_N, C_{RN}), \quad N = R^\perp, \quad \text{Fisher} = J^\top \Sigma^+ J \text{ iff } R \subseteq \text{range}(\Sigma).}$$

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